Exercise path

Start with exercises 1 and 2. Then choose exercise 3 or 4 without help. Exercises 5 to 7 are a bank of transfer, investigation or programming tasks: the teacher assigns one. Exercise 8 is revisited later.

Exercise 6.1 · A collection of derivatives

Consider the functions

$$f(x) = 3x^4 - 2x + 7, \quad g(x) = (x^2 + 2)(x - 1), \quad h(x) = \frac{2x + 1}{x + 3}.$$

1. State the domain of each of these functions.
2. Calculate their derivatives, naming the rule used each time. For g, first keep the two terms of the product rule; for h, write the numerator before simplifying it.
3. Check the result concerning g by expanding its expression before differentiating.

Exercise 6.2 · A complete table

The function f is defined on [0; 4] by $f(x) = x^3 - 6x^2 + 9x + 1$. We are looking for its minimum and maximum values on this interval.

1. Calculate and factorise $f'(x)$, then study its sign on [0; 4].
2. Draw the variation table of f showing the values at the endpoints and at the points where the derivative is zero.
3. Give the minimum and maximum of f on the interval, as well as all the x-coordinates where they are attained.

Exercise 6.3 · A quotient

For every strictly positive real number x, set $f(x) = x + \frac{4}{x}$.

1. Study the variations of f on $]0; +\infty[$. Set out the factors that determine the sign of the derivative.
2. Deduce the smallest value of $x + \frac{4}{x}$, then write an inequality valid for every $x > 0$. State the equality case precisely.

Exercise 6.4 · Compare two curves

The functions $f(x) = x^3$ and $g(x) = x$ are defined on $\mathbb{R}$. Let $\mathcal{C}_f$ and $\mathcal{C}_g$ be their curves in the same coordinate system.

1. Determine the intersection points of these two curves.

2. Study their relative positions on $\mathbb{R}$ by choosing a suitable method. Express your result in terms of the intervals where $\mathcal{C}_f$ is above or below $\mathcal{C}_g$.

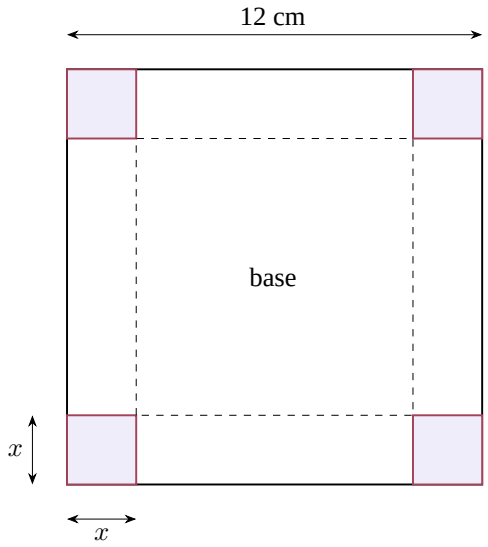
Exercise 6.5 · A package

An open box (no lid) is made from a square sheet of side 12 cm. From each of the four corners we cut out a square of side x cm, then the remaining edges are folded up. The thickness of the sheet is neglected.

1. Draw the net and express the height and the two dimensions of the base of the box. State the values of x which give a box of non-zero volume.

2. Express the volume $V(x)$ in cubic centimetres. Find the dimensions that make this volume maximum.

3. Check the values at both endpoints of the domain and give the exact maximum volume.



The four coloured areas are removed; the dotted lines show the folds. The drawing shows an illustrative value of x and does not give the optimum.

Exercise 6.6 · Stationary without extremum

A pupil claims: "If the derivative is zero at a point, the function must have a local extremum at this point." We study the function $f(x) = x^3$ to test this claim.

1. Calculate $f'(0)$.

2. Compare with $f(0)$ the values of $f(x)$ for real numbers x negative and then positive, as close to 0 a we like.

3. Conclude about the existence of a local extremum at 0 and about the proposed statement.

Exercise 6.7 · Compare Strategies

We want to prove that, for every real number x, $x^2 - 6x + 12 \geq 3$.

1. Give a proof that makes a square appear.
2. Give a second proof by studying the variation of a function.
3. State the equality case. Compare the two methods: which one requires fewer calculations here?

Exercise 6.8 · Delayed review

The reciprocal function $f(x) = \frac{1}{x}$ has domain $D = \mathbb{R} \setminus \{0\}$. A student calculates $f'(x) = -\frac{1}{x^2} < 0$ and concludes that f is decreasing on the whole of D.

1. Identify the missing hypothesis in the application of the theorem linking the sign of the derivative and variations.
2. State the two intervals on which the conclusion is correct.
3. Using the numbers -1 and 1, construct a counterexample to the conclusion on the whole of D.

Independent work. Do only the task assigned at the end of the session, as set in the dated scheme of work: 10-20 min maximum. The exercises depend on the concepts already studied; some questions are therefore postponed. Revisit exercise 8 one week after the chapter (10 min). If the planned time is exceeded, note the difficulty and bring your attempts; do not try to do the whole reserve.

To be done in the CQFD textbook

These exercises are done in your textbook. The scheme of work gives the session; they replace the homework planned for that day (10 to 20 minutes in total).

CQFD, exercise 12 p. 156, questions 1 and 2. Calculate the difference, study its sign and translate this into the relative position of the curves.

CQFD, exercise 16 p. 156, questions f and g. Give, for each function, an interval where it is both negative and increasing; justify both properties separately.